
Maria Teleki

RESEARCH STATEMENT

Socially Robust AI

I build methods, benchmarks, and audits for AI robustness in socially complex situations, at the intersection of Speech AI and Computational Social Science. My primary methodological move is a *diagnose-then-fix loop*: locate where AI commits a social failure – against a specific population (e.g., speakers who stutter) or a social construct from the social sciences (e.g., norms around *who* gets to interrupt *when*) – measure it precisely enough to act on, then design targeted interventions to close the gap. My research connects three threads, each with two short term future directions below and a shared long term direction:

Is AI robust to how people naturally speak? I showed that disfluencies are out-of-distribution for AI models, erasing signals about a speaker (emotion, confidence, etc.), and that the gap has a linguistic structure precise enough to close [P1, P3, P9, P10]. *Future directions*:

- Agents that read power and relationship to interrupt at the right moments [G1]
- Agents that produce disfluency deliberately, to shape what listeners retain

Is AI robust to each person’s uniqueness? I showed that AI is fragile to how a person talks, so identical requests draw different answers and different levels of trust, and that sampling counterfactual personas turns that fragility into a gain [P4, P5, P7, M1]. *Future directions*:

- Evaluation protocols for the $N=1$ user [R1]
- Personas grounded in lived experience via memories

Is AI robust within the sociopolitical structures it operates in? I showed that people and AI read the same policy differently in an audit of 79 U.S. universities [P2]. *Future directions*:

- Audits for the full spectrum of governance interpretation
- Open, continuous governance monitoring at the scale of thousands of institutions

Long term direction: I will build socially situated reasoning: social meaning is settled in context and no dataset can enumerate it, so the fix is reasoning at inference time. I will build methods that perform the social read, and accompanying benchmarks and audits.

My work spans speech venues (ICASSP, Interspeech), sociotechnical venues (FAccT, ICWSM), NLP venues (TACL, ACL, EMNLP Findings, LREC-COLING), and HCI venues (IUI, CHI) with co-authors from Cornell Tech, Google DeepMind, AAAS, KTH, CMU, Edinburgh, KIT, and ETH Zürich.¹ I have also mentored twelve MS and undergraduate researchers, with nearly all co-authoring publications. My PhD is funded in part by the Avilés–Johnson Fellowship. The three threads below share one method, the *diagnose-then-fix loop*, applied to three layers of social context: how a person speaks, who they are, and the structures governing the interaction.

Disfluency-Aware AI: Is AI robust to how people naturally speak?

Voice is fast becoming a primary interface: voice mode in ChatGPT and Claude, voice messages in iMessage and WhatsApp, and more – and all of it runs on real, disfluent speech. I build benchmarks that measure where speech systems fail on natural speech, interventions that close those gaps, and systems that reason about disfluency directly. Disfluencies like “uh,” “um,” “well,” “I guess,” and mid-sentence restarts (“we should – actually, let’s wait”) are not just noise, they carry social signal: different genders use disfluencies differently (ICWSM ’25 [P7]), and well-placed disfluencies can be used to

¹Citations (e.g., [P1]) refer to publications listed in my CV.

shape what listeners remember.² They also degrade downstream performance, from spoken-content summarization (LREC-COLING '24 [P10]) to conversational recommendation (Interspeech '25 [P6]). Yet, automatic speech recognition (ASR) systems systematically fail to transcribe³ disfluencies (Interspeech '24 [P9]), so they rarely reach the corpora that LLMs and SpeechLLMs learn from – making them out-of-distribution signals. As shown in Figure 1, dropping disfluencies has consequences in high-stakes settings: in medical translation, it removes cues a doctor uses to gauge a patient. To fill the gap in training data, the field usually reinjects disfluencies at random, treating as noise what carries dense, context-dependent social meaning. My work starts from their linguistic structure.

I operationalized Shriberg's disfluency taxonomy³ – which consists of categories such as interjections (*uh, um*), parentheticals (*I mean*), and edited repairs (*Target omg sorry Walmart*) – as a metric, which we call $\mathcal{Z}$ -Scores (ICASSP '26 [P3]), which scores how completely a system removes each category. On a benchmark of real phone conversations, $\mathcal{Z}$ -Scores pinpointed which disfluency categories a model handled worst; I designed a targeted prompt to address them, and those categories improved – a diagnose-then-fix loop that previous disfluency removal metrics can't support. I then asked: if one model has a distinctive failure profile, do others? With a collaborator at KTH, I built an evaluation suite that compares speech-capable LLMs (In Prep. for ARR [R4]). We find that each model has a consistent editing style set by its training, and reasoning-capable models over-delete, removing fluent content.



Figure 1: Smart glasses translate a patient's speech in real time. Because disfluencies signal a patient's confidence, translation that drops them loses clinically meaningful information, the gap our Uh-Mazing benchmark measures [P1].

With collaborators at KIT, ETH Zürich, and Carnegie Mellon, I built Uh-Mazing, a multilingual benchmark for disfluency-aware speech translation (Figure 1), to be released via Penn LDC (Conditionally Accepted at TACL [P1]). We find that false starts and self-repairs drive most of the translation-quality loss, and failed disfluencies are usually dropped outright – a loss that can shift the perceived emotion by up to 71%, meaning the speaker's feelings get misread. Inference-time methods narrow this gap.

At my Google DeepMind internship, I built social reasoning for interruptions (In Prep. for FAccT [G1]): reasoning over the paralinguistic signals of speech (e.g., hesitation, volume) and disfluencies alongside the social dynamics of power, relationship, and culture to determine the meaning of an interruption, toward building systems that can interrupt appropriately.

These measurements assume one correct transcript exists. It doesn't: with collaborators at Cornell Tech, I introduced the Linguistic Fidelity Spectrum (In Prep. for CSCW [R2]), a framework to characterize the range of valid transcriptions, from word-for-word to cleaned-up. Our follow-on work addresses the epistemic injustice of single-ground-truth setups (In Prep. for ARR [R3]).

Short Term Direction 1: The field frames interruption as a noise injection problem. Following our cross-community agenda [R1], the missing layer is *social*: the same pause can invite interruption from a peer and discourage it from a subordinate. I will extend the interruption reasoning from my DeepMind project [G1], so proactive agents in meetings can participate without derailing the people they serve.

²Diachek & Brown-Schmidt, 2023, *The Effect of Disfluency on Memory for What Was Said*

³Listeners filter disfluencies out of perception automatically, making them labor-intensive to annotate faithfully (Shriberg, 1994, *Preliminaries to a Theory of Speech Disfluencies*).

Short Term Direction 2: My work so far asks whether systems are robust to *human* disfluency; the inverse is whether *systems* should produce it. Disfluencies shape listener memory and attention – a hesitation before a word makes it more memorable² – so a speaking agent can place them deliberately to emphasize information, flag uncertainty, or slow a listener down. I will design a method to generate communicative disfluency and measure what listeners retain, trust, and understand, turning disfluency into a signal that systems can use to better communicate with users.

Personalized AI: Is AI robust to each person's uniqueness?

Voice assistants, tutors, and agents are all racing to personalize. But adapting to a person first requires understanding them, and AI understands some people better than others: my work measures where that understanding breaks down, then closes the gap. I showed that discourse style alone shapes how well a person is represented: identical information is served unequally depending on speaking style (ICWSM '25 [P7]). In SpeechLLMs the problem deepens, because voice and identity compound (Interspeech '26 [P4]), and the harm is causal: voice conversion experiments show a speaker's voice directly shapes the trust, attribution, and empathy they receive (IUI Poster '26 [M1]). In CHOIR, we show that individual differences are a resource: counterfactual personas each excel on different parts of a task, and composing them lifts reasoning accuracy up to 20% (ACL '26 [P5]).

Short Term Direction 1: AI evaluation has two strong traditions: aggregate metrics that scale, and studies of individual experience that go deep. Combining them opens a new kind of evaluation for highly intersectional users, such as someone with a unique set of health conditions. Drawing on our cross-community agenda [R1], I will build evaluation protocols in which $N=1$ users decide what the system should get right – e.g., one person may want an AI system to flag every possible medication interaction, while another finds that overwhelming and wants only what is urgent. Mixed methods then measure the system, pairing aggregate metrics with the user's criteria. The result is a protocol that tests whether a system can adapt to $N=1$ users, even when their needs conflict.

Short Term Direction 2: When we build personas, we reduce a person to a few neat attributes (e.g., gender, race, age), but how someone reads a situation comes largely from a lifetime of accumulated context. Building on my storytelling survey (EMNLP Findings '25 [P8]), memories in great literary works often come back in flashes of relevance, e.g., *At four, I fell off that swing in our backyard because I was staring at a flower, and it felt like slow motion. I remember my arm snapping when I hit the grass. My dad was so panicked, he switched to Hungarian, and couldn't switch back until we were halfway to the ER.* I will build memory-grounded models of lived experience, extending my persona work from the attributes a system assigns toward the history a person carries.

Governable AI: Is AI robust within the sociopolitical structures it operates in?

Governance is a moving target: what a policy means is situated, depending on who is affected and who enforces it. Auditing even one institution means AI reasoning jointly over policy documents, appeals procedures, and more; keeping that current across a sector is a standing computational commitment. Our FAccT '26 work [P2] is the first instantiation. ACAI-US79 tests whether people and AI read policy the same way. A human panel and an LLM judge independently evaluate every policy, with the human panel as ground truth. Disagreement is the measurement: (i) where the human panel diverges, the policy is ambiguous, and (ii) where the LLM judge diverges from the human panel, the LLM judge is unreliable. Evaluations populate a scorecard over four dimensions (policy clarity, faculty support, feedback mechanisms, detection-tool governance) that aggregate into ACAI_u, an interpretable score in $[0, 100]$. I released the whole system – dataset, audit instrument, and a public site at acai-us79.org – so institutions can inspect their scores and others can extend the audit.

Short Term Direction 1: No rule has a single reading: in ACAI-US79, the human–LLM divergence was the finding. I will build audits that model the full spectrum of interpretation: a diverse population of personas reading each rule, using the persona infrastructure from my personalization work.

Short Term Direction 2: Rules decay by drift and by design, and today they are fixed only when someone can afford the fight. ACAI-US79 is large, covering 79 universities, but democratic life is larger still, covering thousands of institutions. I will build open, continuous monitoring at that scale, so anyone can verify the auditor and run counteraudits.

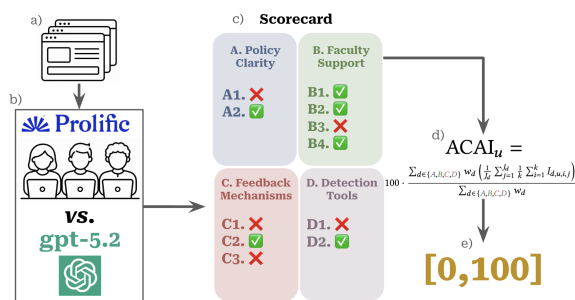


Figure 2: The ACAI-US79 pipeline: policies from 79 U.S. universities (a) are evaluated by a human panel and an LLM judge (b), scored across four dimensions (c), and aggregated into $ACAI_u \in [0, 100]$ (d–e).

Long Term Direction: Toward Socially Situated Reasoning

My three threads converge on one hypothesis: just as meaning is situated in a social context, AI interaction must be situated in inference-time reasoning.⁴ What a word means, what a silence permits, what a policy requires is settled in context – *by who is speaking, to whom, with what lived experience, in what power dynamic, under which societal structures, at which point in time, and more*. Because this space of configurations is combinatorially too large for any dataset to enumerate, the social read has to happen at inference time via reasoning.

That reasoning combines my three threads: how a person speaks, who they are, and the structures governing the interaction. My surveys of inference-time methods [M8] and of narratives [P8] map a maturing toolkit for it, one that can extend to other modalities. My Google DeepMind work is the first system in this mold, reasoning jointly over paralinguistic signals and social dynamics to decide when interrupting is appropriate [G1].

My agenda generalizes this pattern into a research program: benchmarks that score whether a system socially reads the situation correctly, methods that perform the reading, and audits that hold deployed systems accountable for it. While we should resist having our human lives reduced to a few neat social attributes, we also deserve to be understood and served well by AI [R1]. Finding that middle ground, and moving the field toward it, is my long term agenda.

Funding Landscape. I will aim for these funding opportunities: *Disfluency-Aware AI* and *Personalized AI* fit NSF CISE Robust Intelligence, Human-Centered Computing, and Information Integration and Informatics, as well as NSF SaTC 2.0, and Google and Amazon industry awards, and calls from the Cooperative AI Foundation (CAIF). *Disfluency-Aware AI* also fits the NIH; *Governable AI* fits the Knight Foundation and the Mozilla Foundation.

Students and Lab. Robust social reasoning is interdisciplinary, and so is the lab I want to build. My problems draw on CS/EE, information science, linguistics, psychology, and law. My pipeline moves students from contributing to my projects toward leading their own (detailed more in my Teaching Statement) [M2, M3, M4, M9] (3 UG, 1 MS); two projects earned first-place Student Research Week awards (1 UG, 1 MS); one student completed an REU at RIT and three interned at Google, Meta, and Target this summer (UGs); a graduated undergraduate just started at Cornell Tech and graduated MS students have gone to Amazon and NVIDIA. I target venues across speech AI, computational social science, and NLP, making my lab a natural fit for many different types of students.

⁴The situated-action tradition (Suchman, 1987, *Plans and Situated Actions*).